

flea NEWS 41



Department of Entomology Iowa State University Ames, Iowa 50011-3222

10th ANNIVERSARY ISSUE

Welcome to the 41st issue of **Flea News**. It has been a busy six months for us, and the holiday season is now upon us. As a result we have not done the usual library search that normally precedes the production of one of these issues. Still, we feel that there is enough information to justify keeping to our usual timetable of twice annual issues. We will try to take up the slack in the next number.

NOTICES:

E. W. Classey Ltd., Bookseller, has announced that, in addition to their regular catalogues, they have available a series of special subject lists. As of the past summer they had printed lists for the Diptera, Hymenoptera, Coleoptera, Lepidoptera and Botany. Interested readers may write the firm at P. O. Box 93, Faringdon, Oxon SN7 7DR, ENGLAND, or telephone 036782 399. Fax: 036782 429.

In addition to our regular mailing address the editors of **Flea News** may now be contacted at the following addresses:

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NEW CURATOR We have recently been informed that EVERT LINDQUIST is now

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nominally responsible for the Siphonaptera collection at the Canadian National Collection. He can be contacted at: Agriculture Canada, Biosystematics Research Centre, K. W. Neatby Building, C. E. F., Ottawa K1A 0C6, Ontario, Canada.

MORE FLEA COLONIES

A new subscriber to **Flea News**, Liu Jun, has written that he has established colonies of *Nosopsyllus laeviceps kuzenkovii*, *Frontopsylla luculenta*, *Amphipsylla primaris* and *Neopsylla bidentatiformis*. Readers wishing to correspond will find his address in the Mailing List section.

CONGRATULATIONS

Michael Dryden has received his Ph. D. degree from Purdue University. His dissertation is entitled *Blood consumption and feeding behavior of the cat flea, Ctenocephalides felis felis (Bouché, 1835)*. He was recently spotted on the "Good Morning, America" TV program participating in a discussion of ultrasonic flea collars.

D E C E M B E R

1990

JUST FOR STAMP COLLECTORS. Three hundred years ago, postal authorities started to decontaminate letters coming from plague infested areas. This practice continued even as late as 1850 when vinegar baths and juniper smoke were used to purify envelopes from disease stricken sections of the world. These envelopes can be detected by small pin holes and special decontamination markings and are considered a highly collectible item today.

When adhesive postage stamps were first issued, many people feared that licking the gum on the back would transmit some form of disease. After several studies, it was shown that diseases or a virus could not be transmitted via postage stamp gum. (from **The [Ames] Daily Tribune**, 28-October-1989.)



RECENTLY PUBLISHED

The Ceratophyllidae: currently accepted valid taxa (Insecta: Siphonaptera). R. E. Lewis. 1990. ISBN 3-87429-304-1. Koeltz Scientific Books, D-6240 Koenigstein, Germany. 267 pp., 1 fig. (unnumbered). DM 150.00, US\$ ca. 96.00. (Available in the United States through Lubrecht & Cramer, Ltd., R. D. #1, Box 244, Forestburgh, NY 12777.)

The Ceratophyllidae is the only siphonapteran family that has not been monographed in the Catalogue series published by the British Museum (Natural History). Although the 1983 volume by Traub, Rothschild and Haddow concerned itself with various aspects of this family, it is totally devoid of nomenclatural considerations beyond the erection of seven new genera, eight new subgenera and the construction of a key to the genera and subgenera of the world fauna by Smit. *The Ceratophyllidae* is an attempt to provide the bibliographic and nomenclatural information about the 515 species and subspecies in a single volume.

After a brief introduction, a materials and methods section deals with types, collecting localities, host identifications and journal abbreviations. This is followed by a history of the higher classification of the family. There follows 152 pages of a systematic inventory of what are currently considered the valid species and subspecies in the family up to 31-December-1989. Information about each taxon is arranged in the following order: name, author, date of description, citation of the original description, collection locality data in order of decreasing area (i.e., country, state, province, county, locality), map coordinates if available, type host, collection date and collector. This is followed by information about type deposition, if known. Similar information is given for junior synonyms, including where, when and by whom they were synonymized. This section is arranged alphabetically by genus, subgenus, species and subspecies. Acknowledgments are followed by a literature-cited section containing 460

references. Appendix I lists the type depositories and their abbreviations as cited in the text. Appendix II lists the names of all of the avian and mammalian hosts and the citations of their original descriptions. Appendix III lists the abbreviations of all of the scientific periodicals cited in the text with their expanded equivalents. There are 37 pages of index in which the species of fleas and their hosts are cross-indexed.

It is my contention that the obscurity of information such as is included in this volume has done much to discourage attention to this family by beginning students of the order. Before a satisfactory understanding of the family can be attained, a number of existing genera must be rigorously revised and many new genera described. The Ceratophyllidae, perhaps more than any other of the 15 siphonapteran families, continues to be in dire need of attention by specialists.

R.E.L.

CONFERENCE NOTES

The annual meeting of the Entomological Society of America was held December 2-6, 1990, at New Orleans, Louisiana. Papers presented relating to fleas are listed below. The mailing address of the presenter is supplied.

Blood digestion by adult cat fleas: implications for parental care. J. Silverman (American Cyanamid, Shulton Research Division, 697 Route 46, Clifton, NJ 07015) & A. G. Appel.

Effect of host activity on egg drop by cat fleas, *Ctenocephalides felis*. W. H. Kern, Jr. (Department of Entomology & Nematology, University of Florida, Gainesville, FL 32611), N. C. Hinkle, P. G. Koehler & R. S. Patterson.

Mode of action of pyriproxyfen and methoprene on cat flea (*C. felis*) eggs. K. G. Palma (Department of Entomology, Texas A & M University, College Station, TX 77843), S. M. Meola & R. W. Meola.

The influence of relative humidity on the toxicity of insecticides against cat fleas, *Ctenocephalides felis*. M. K. Rust (Department of Entomology, University of California, Riverside, CA 92521) & D. A. Reiersen.

Residual efficacy of three insect growth regulators as premise treatments for control of cat fleas. N. C. Hinkle (Department of Entomology & Nematology, University of Florida, Gainesville, FL 32611), P. G. Koehler & R. S. Patterson.

Preparation of nonsclerotized insect eggs for light microscopy. S. M. Meola
(USDA-ARS, FAPRL, Route 5, Box 810, College Station, TX 77840), K.
G. Palma & R. W. Meola.

REPRINTS RECEIVED

The following readers have sent reprints and other literature since Flea News 40. Thank you. We are very grateful. A. F. Azad, A. H. Benton, A. Dudich, T. Galloway, A. I. Goncharov, C. Guiguen, B. Kotty, J. Kucera, R. Lane, S. N. Rybin, V. F. Sapegina, D. Striese and Xie Bao-qi.

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DELETION

David W. Lupton

A PREFATORY NOTE TO THE SPECIES LIST - In keeping with Mr.
Smit's practice of numbering the taxa chronologically, the following list
includes numbers for those taxa described in 1988. Readers are urged to
call any omissions in these lists to our attention.

- 98801 apiciflata Liu, Xu & Li Amphipsylla
- 98802 aculeolatus Ge & Ma Paradoxopsyllus
- 98803 yadongensis Wang & Wang Amphipsylla

- 98804 *angustus* Beaucournu & Launay *Nosopsyllus* (*Gerbillophilus*)
maurus
 98805 *garamanticus* Beaucournu & Launay *Nosopsyllus*
 (*Gerbillophilus*)
 98806 *incisus* Beaucournu & Torres-Mura *Delostichus*
 98807 *lewisi* Beaucournu & Gallardo *Plocopsylla*
 98808 *propinqua* Beaucournu & Gallardo *Ctenoparia*
 98809 *liui* Wu, Chen & Liu *Nycteridopsylla*
 98810 *arcuata* Ge, Wang & Ma *Callopsylla* (*Callopsylla*)
 98811 *novaehiberniae* Beaucournu & Mahnert *Metastivalius*
 98812 *sagitta* Yu, Ye & Liu *Mesopsylla*
 98813 *infratentus* Wu, Wang & Liu *Ischnopsyllus* (*Hexactenopsylla*)
 98814 *longispina* Zhang & Yu *Callopsylla* (*Callopsylla*)
 98815 *aenigma* Lewis & Cheetham *Pulicella*
 98816 *kasogonaga* Schramm & Lewis *Plocopsylla*
 98817 *angustihamulus* Li, Zhang & Zeng *Macrostylophora*
 98818 *longiprojiciens* Chen, Li & Wei *Ctenophthalmus*
 (*Sinoctenophthalmus*)
 98819 *formozovi* Darskaya *Coptopsylla lamellifer*
 98820 *angustisinus* Cai & Wu *Paradoxopsyllus*
 98821 *lui* Gong & Lei *Cratynius* (*Angustus*)
 98822 *sintrensis* Ribeiro & Capela *Typhloceras favosum*

PROVISIONAL LISTS OF TAXA DESCRIBED IN 1989 AND 1990

1989

- Palaeopsylla laxidigita* Xie & Gong
Palaeopsylla polypina Xie & Gong
Palaeopsylla medimina Xie & Gong
Ctenophthalmus (*Palaeoctenophthalmus*) *harputus* Aktaş
Ctenophthalmus (*Ctenophthalmus*) *solutus nasutus* Beaucournu, Valle &
 Launay
Citellophilus simplex rosickyi Cyprich
Xenopsylla guancha Beaucournu, Alcover & Launay
Stenischia chini Xie & Lin
Stenischia liae Xie & Lin
Stenischia liui Xie & Lin
Stenischia wui Xie & Lin
Jellisonia amadoi Ponce-Ulloa
Jellisonia mexicana Ponce-Ulloa
Plocopsylla crypta Beaucournu & Gallardo
Rowleyella gongshanensis Gong & Duan

1990

Hectopsylla pascuali Beaucournu & Alcover
 Plocopsylla consobrina Beaucournu & Alcover
 Agastopsylla boxi gibbosa Beaucournu & Alcover
 Chiliopsylla allophyla tonnii Beaucournu & Alcover
 Rhadinopsylla (Sinorhadinopsylla) leii Xie, Gong & Duan
 Smitipsylla quadrata Xie & Li
 Rowleyella nujiangensis Lin & Xie
 Hystrichopsylla llorentei Ayala & Morales
 Jellisonia guerrerensis Morales
 Ctenoparia intermedia Beaucournu & Kelt
 Plocopsylla wilesi Beaucournu & Kelt
 Plocopsylla silewi Beaucournu & Kelt
 Tiamastus gallardoi Beaucournu & Kelt
 Nosopsyllus (Nosopsyllus) wualis rongjiangensis Li & Huang
 Nosopsyllus (Nosopsyllus) wualis bosensis Li & Pan
 Nosopsyllus (Nosopsyllus) wualis diandongensis Li, Xie & Yang
 Nosopsyllus (Nosopsyllus) elongatus longchuanensis Li, Xie & Wu
 Nosopsyllus (Nosopsyllus) elongatus puerensis Li, Xie & Yu
 Nosopsyllus (Nosopsyllus) elongatus yanshanensis Li, Xie & Liao

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(1979-1985)

LITERATURE ON SIPHONAPTERA

1979 (List 21)

Zapatero, L. M., J. L. Guillen & J. Gil Collado. Estudio parasitologico de una colonia *Tadarida taeniotis*. Com. II Cong. Nac. Parasitol., Léon.

1982 (List 14)

Zimmerli, J. Étude des parasites de la fouine (*Martes foina*) dans le canton de Vaus durant la période 1980-1981. Schweizer Archiv für Tierheilkunde 124: 419-422.

1985 (List 11)

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Pugh, R. E. & D. E. Moorhouse. Factors affecting the development of *Dipylidium caninum* in *Ctenocephalides felis felis* (Bouché, 1835). Zeitschrift für Parasitenkunde 71(6): 765-775.

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1986 (List 9)

Jimenez, F. & F. J. Marquez. Biometria de *Caenopsylla laptevi ibera* Beaucournu et Marquez. Actas de las VIII jordanas A et E, Sevilla, Oct. 1986, 1.157-1.166.

Zarzara, C. Data on parasitoses in furbearing animals: aetiology and control. Revista de Cresterea Animalelor 36(11): 30-34.

1987 (List 7)

Büchi, R. Frequently occurring pests of dwellings and farms. Mitteilungen der Schweizer Landwirtschaft 35(5): 131-144.

Stehlík, M. Entomology. Pp. 51-57 in: *Companion Bird Medicine*, Burr, E. W. (Ed.). Iowa State University Press, Ames. v-x + 247 pp. [Siph.: 55-56, fig. of *E. gallinacea*]

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Costantini, R., P. Lanfranchi, A. Iori, M. A. Abdullatif & T. Balbo. Contributo allo studio dei sifonatteri dela Somalia. Parassitologia 30: 203-209.

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- Dudich, A.** Notes on the interspecific hybrids of fleas (Siphonaptera) living on small mammals. Programme & Abstracts of Vth EMOP (European Multicolloquium of Parasitology). September 4-9, 1988, Budapest, Hungary: 4-33.
- El-Gazzar, L. M., R. S. Patterson & P. G. Koehler.** Comparisons of cat flea (Siphonaptera: Pulicidae) adult and larval susceptibility. Florida Entomologist 71(3): 359-363.
- Fukase, T. & H. Itagaki.** Efficacy of a 0.1% propoxur shampoo product against fleas on dogs. Journal of the Japanese Veterinary Medical Association 41(12): 851-855.
- Guiguen, C.** Anthropozoonoses et oiseaux marins: Contribution à l'étude des ectoparasites hématophages des espèces nicheuses sur les côtes françaises continentales et insulaires. Thèse. Université d'Aix - Marseille II. 326 pp.
- LeRiche, P. D., A. K. Sol, Q. Alemzada & L. Sharifi.** Parasites of dogs in Kabul, Afghanistan. British Veterinary Journal 144(4): 370-373.
- Li W.-d. & Yu X.** A new record of *Ischnopsyllus* from China. Acta Zootaxonomia Sinica 13(1): 80. [*I. octactenus* Kolenati.]
- Navephap, S.** A survey of the ectoparasites in the domestic rabbit. Thai Journal of Veterinary Medicine 18(1): 25-31.
- Nihad, W., ? Al-Khalidi, I. Tafiq, ? Al-Alousi & A. H. Subber.** Internal and external parasites in cats in Mosul, Iraq. Journal of Veterinary Parasitology 2(2): 137-138.
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- Raja, E. E. & M. Anantaraman.** On the occurrence of the porcupine flea, *Pariodontis riggenbachii wernecki* Costa Lima, 1940, in Tamil Nadu. Cheiron 17(3): 128-129.
- Sagdieva, P. D., Sh. G. Tsikhistavi & N. A. Nikulina.** The ectoparasites of small rodents in the Great Caucasus. Soobshcheniya Akademii Nauk Gruzinskoi SSR 129(1): 165-168.

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Southcott, R. V. Some harmful Australian insects. Medical Journal of Australia 149(11/12): 656-662.

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Dudich, A. Náčrt fauny drobných cicavcov (Insectivora, Rodentia, Chiroptera) okresu Trnava (Malé Karpaty, Trnavská pahorkatina). Zborník odborných prác VI. Západoslovenského TOP-u, Zväzok V, Buková, Bratislava: 92-105.

Fukase, T. & H. Itagaki. Efficacy of propoxure against fleas on domestic cats. Journal of the Japanese Veterinary Medical Association 42(7): 463-467.

Faulkner, C. T., S. Patton & S. S. Johnson. Prehistoric parasitism in Tennessee: evidence from the analysis of desiccated fecal material collected from Big Bone Cave, Van Buren County, Tennessee. Journal of Parasitology 75(3): 461-463.

Grant, D. I. Parasitic skin diseases in cats. Journal of Small Animal Practice 30(4): 250-254.

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Striese, D. Beitrag zur Kenntniss der Siphonaptera der Oberlausitz. Abhandlungen und Berichte des Naturkundemuseums Görlitz 63(6): 1-11.

Tonelli, E. Therapeutic use of limonene, piperonyl butoxide and cypermethrin given as a bath for ectoparasitoses in dogs and cats. Veterinaria Argentina 6(57): 456-459.

Vakhrousheva, Z. P., A. D. Gorchakov, N. A. Kolupaeva & E. G. Chernykh. The use of flannel flags for collecting fleas at the entrance of burrows of steppe rodents. Meditinskaya Parazitologia i parazitarnye bolezni (1): 54-57.

Varshavskiy, S. N., M. N. Shilov, N. V. Popov, A. V. Survillo, M. A. Tarsov, V. P. Kozakevitch, P. S. Denisov, B. S. Varshavskiy, Z. S. Sorokina, A. O. Adamyan, P. D. Golubev & P. N. Korzhov. Distribution of *Rattus norvegicus* in the south-eastern European part of the USSR and tasks of the anti-plague service. Zoologicheskii Zhurnal 68(10): 85-94.

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Werner, G., T. Hiepe, P. Tschauschew, H. Hahn, H. Zimmermann, W. Siering & L. Hoffmann. Vioxan powder - a new dry preparation for use against ectoparasites of domestic animals. Monatshefte Veterinarmedizin 44(17): 611-613.

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cat fleas, (*Ctenocephalides felis* (Bouché)), with a rickettsia-like microorganism. American Journal of Tropical Medicine and Hygiene 43(4): 400-409.

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Beale, K. M., G. A. Kunkle, L. Chalker & R. Cannon. Effects of sedation on intradermal skin testing in flea-allergic dogs. Journal of the American Veterinary Medical Association 197(7): 861-864.

Beard, C. B., J. F. Butler & D. W. Hall. Prevalence and biology of endosymbionts of fleas (Siphonaptera: Pulicidae) from dogs and cats in Alachua county, Florida. Journal of Medical Entomology 27(6): 1050-1061.

Beaucournu, J. C. & D. A. Kelt. Contribution a la faune du Chili: puces nouvelles ou peu connues de la partie sud (Insecta: Siphonaptera). Revue Suisse de Zoologie 97(3): 647-668. [*Ctenoparia intermedia*, *Plocopsylla wilesi*, *P. silewi*, *Tiamastus gallardoi*, female of *Listronius ulus*]

Best, T. L., A. S. Titus, K. Caesar & C. L. Lewis. *Ammospermophilus harrisii*. Mammalian Species 366: 1-7 [Siph.: 4]

Best, T. L., A. S. Titus, C. L. Lewis & K. Caesar. *Ammospermophilus nelsoni*. Mammalian Species 367: 1-7 [Siph.: 5]

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- Dryden, M. W.** Blood consumption and feeding behavior of the cat flea, *Ctenocephalides felis felis* (Bouché, 1835). PhD. Dissertation. Purdue University, West Lafayette, IN. 128 pp.
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It will be understood that - although their titles are given in English in these lists - papers in Russian or Chinese periodicals, or books published in the USSR or China, are in Russian or Chinese (but often with summaries in English).

Flea News is a twice annual newsletter devoted to matters involving insects belonging to the order Siphonaptera. It is compiled and distributed free of charge by Robert E. and Joanne H. Lewis with support from the Department of Entomology at Iowa State University in Ames, Iowa, USA. It is mainly bibliographic in nature, but recipients are urged to check any citations cited here before including them in a publication. Many of our sources are abstracting journals and current literature sources such as **Current Contents®** and citations have not necessarily been checked for accuracy. Further, recipients are urged to contribute items of interest to the profession for inclusion herein. -R.E.L.

The editors of **Flea News** wish to take this opportunity to wish our subscribers a Very Merry Christmas and we extend our best wishes to you all in the year to come. - R.E.L. & J.H.L.

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